THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET A

Full Syllabus (Chapters 9–14) | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The converse of "If n is even, then n^2 is even" is "If n^2 is even, then n is even". Which statement is correct? 1

(a) both the proposition and its converse are true
(b) only the proposition is true
(c) only the converse is true
(d) both are false

2. Which of the following is a counterexample to the proposition "If n is a prime number, then $n + 2$ is also a prime number."? 1

(a) 5 **(b)** 11 **(c)** 3 **(d)** 7

3. A group of 20 students has an average of 55 marks and another group of 30 students has an average of 65 marks. The average of all 50 students is: 1

(a) 63 **(b)** 61 **(c)** 60 **(d)** 62

4. 6 units of a mixture with 4% metal is mixed with a mixture of 10% metal to get 8% metal. The quantity of the second mixture needed is (same units): 1

(a) 24 **(b)** 52 **(c)** 32 **(d)** 12

5. In the column-addition algorithm, the basic step we assume we can do directly is: 1

(a) adding two single-digit numbers **(b)** counting dots one by one
(c) adding two 10-digit numbers **(d)** dividing by 10

6. The list of divisors of 18 produced by the divisor algorithm has how many entries? 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

7. To be called a quadrilateral ABCD, the four distinct points A, B, C, D must have: 1

(a) no three collinear, all in one plane, and non-self-intersecting sides
(b) points in three different planes
(c) three points collinear
(d) all four points on one line

8. In △ABC, P and Q are the midpoints of AB and AC and BC = 18 cm. The length of PQ is: 1

(a) 18 cm **(b)** 36 cm **(c)** 9 cm **(d)** 11 cm

9. The slope of the line through the points (0, -3) and (4, 5) is: 1

(a) 1/2 **(b)** 3 **(c)** 2 **(d)** -2

10. Which of the following points lies on the line 4x + 3y = 10? 1

(a) (2, 3) **(b)** (1, 3) **(c)** (2, 2) **(d)** (1, 2)

11. The total surface area of a cuboid of length 10 cm, width 6 cm and height 4 cm is: 1

(a) 240 cm^2 **(b)** 134 cm^2 **(c)** 248 cm^2 **(d)** 124 cm^2

12. Taking $\pi = \frac{22}{7}$, the curved surface area of a cone of base radius 7 cm and slant height 25 cm is: 1

(a) 704 cm^2 **(b)** 275 cm^2 **(c)** 1100 cm^2 **(d)** 550 cm^2

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. State the formula for the combined average when collections of sizes n and m have averages a and b. 2

14. How many numbers must be checked to compute divisors(n) by the algorithm of the chapter? Why can the list of divisors be in increasing order? 2

15. Why must no three vertices of a quadrilateral ABCD be collinear? What shape do we get if three are collinear? 2

16. Why does the line $5x - 2y = 0$ pass through the origin? 2

17. Why does the formula for the volume of a cone contain the factor $\frac{1}{3}$? How can this be shown by an experiment? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Prove: if d divides both m and n (m ≥ n), then d divides m − n. 3

19. The diagonal AC of parallelogram ABCD bisects $\angle A$. Show that it also bisects $\angle C$. 3

20. A line passes through $(1, 3)$ and $(3, 7)$. Find its slope and its equation in the form $y = mx + d$. 3

21. A pyramid has a square base of side 9 cm and height 10 cm. Find its volume. How does it compare with a prism of the same base and height? 3

22. The angles of a quadrilateral are in the ratio 2 : 3 : 3 : 4. Find each angle. 3

OR

Prove that a quadrilateral with one pair of opposite sides equal and parallel is a parallelogram. 3

23. Solve graphically: $x + y = 6$ and $x - y = 2$. Write a table of two points for each line and state the nature of the solution. 3

OR

The sum of the digits of a two-digit number is 11. The number is 27 more than the number formed by reversing its digits. Find the number. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** At a stall, the pani has 8% spice. Customers complain. The owner wants to reduce the spice to $\frac{3}{4}$ of the original concentration. 5

(i) What is the new concentration?

(ii) How much water must be added to 10 litres?

(iii) What is the total volume after adding?

OR

Case study: A cricket academy has 9 seniors with average height 168 cm and 6 juniors with average height 152 cm. 5

(i) Find the sum of the heights of the seniors and of the juniors.

(ii) Find the average height of all 15.

(iii) Why does $\frac{168+152}{2}$ give a wrong result?

25. **Case study:** Three identical metal spheres, each of radius 3.5 cm, fit exactly in a row inside a closed cuboidal box. 5

(i) Write the internal dimensions of the box.

(ii) Find the volume of the box.

(iii) Find the total volume of the three spheres.

(iv) The remaining space is filled with preservative liquid; find its volume ($\pi = \frac{22}{7}$).

OR

Derive the formulas for the curved surface area and total surface area of a cylinder of radius r and height h. Hence find both for $r = 7$ cm and $h = 15$ cm ($\pi = \frac{22}{7}$). 5

END OF QUESTION PAPER — Total: 50 Marks